

# **EC200U&EG91xU Series**

## **QuecOpen(SDK) Serial Port API Reference Manual**

**LTE Standard Module Series**

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# About the Document

## History

| Revision | Date       | Author    | Description                                                                                                                                                                                                                                                                                                                                                                          |
|----------|------------|-----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| -        | 2021-10-25 | Braden HE | Creation of the document                                                                                                                                                                                                                                                                                                                                                             |
| 1.0      | 2021-10-25 | Braden HE | First official release                                                                                                                                                                                                                                                                                                                                                               |
| 1.1      | 2023-09-21 | Sum LI    | <ol style="list-style-type: none"><li>1. Updated the document name based on the unified QuecOpen naming.</li><li>2. Added applicable modules EG912U-GL and EG915U series.</li><li>3. Updated the enumeration of baud rate (Chapter 2.3.1.4)</li><li>4. Updated the log tool description and added a note about the log tools supported by different modules (Chapter 3.2).</li></ol> |

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# 1 Introduction

Quectel EC200U series and EG91xU family modules support QuecOpen® solution. QuecOpen® is an embedded development platform based on RTOS system, which is intended to simplify the design and development of IoT applications. For more information on QuecOpen®, see **document [1]**.

This document is applicable to QuecOpen® solution based on CSDK build environment. It introduces the serial port API functions, related examples and feature debugging of EC200U series and EG91xU family modules. The modules support the software to configure 3 physical serial ports and 3 USB virtual serial ports. The RX of the physical serial port uses the DMA mode by default, and the TX uses the FIFO mode by default.

## 1.1. Applicable Modules

**Table 1: Applicable Modules**

| Module Family | Module        |
|---------------|---------------|
| -             | EC200U Series |
| EG91xU        | EG912U-GL     |
|               | EG915U Series |

## 2 Serial Port API

### 2.1. Header File

*ql\_uart.h*, the header file of serial port API, is in the *components\ql-kernel\in* directory in SDK. Unless otherwise specified, all the header files mentioned in this document are in this directory.

### 2.2. API Overview

**Table 2: API Overview**

| Function                       | Description                                             |
|--------------------------------|---------------------------------------------------------|
| <i>ql_uart_set_dcbconfig()</i> | Configures the attributes of serial port.               |
| <i>ql_uart_get_dcbconfig()</i> | Gets the attribute configuration of serial port.        |
| <i>ql_uart_open()</i>          | Opens the specified serial port.                        |
| <i>ql_uart_close()</i>         | Closes the specified serial port.                       |
| <i>ql_uart_write()</i>         | Writes data to the specified serial port.               |
| <i>ql_uart_read()</i>          | Reads the data from the specified serial port.          |
| <i>ql_uart_register_cb()</i>   | Registers the callback function for serial port events. |



## 2.3. API Description

### 2.3.1. ql\_uart\_set\_dcbconfig

This function configures the attributes of serial port. The parameters configured by this function take effect after calling `ql_uart_open()` to reopen the serial port.

- **Prototype**

```
ql_uart_errcode_e ql_uart_set_dcbconfig(ql_uart_port_number_e port, ql_uart_config_s *dcb)
```

- **Parameter**

*port:*

[In] Serial port number. See **Chapter 2.3.1.1** for details.

*dcb:*

[In] Serial port attribute configuration. See **Chapter 2.3.1.2** for details.

- **Return Value**

See **Chapter 2.3.1.3** for details.

#### 2.3.1.1. ql\_uart\_port\_number\_e

Currently, 3 physical serial ports and 3 USB virtual serial ports are supported by the module. The enumeration of the serial port number is defined as follows:

```
typedef enum
{
    QL_PORT_NONE = -1,
    QL_UART_PORT_1,
    QL_UART_PORT_2,
    QL_UART_PORT_3,
    QL_USB_PORT_AT,
    QL_USB_PORT_MODEM,
    QL_USB_PORT_NMEA,
    QL_PORT_MAX,
}ql_uart_port_number_e
```

- Member

| Member                         | Description     |
|--------------------------------|-----------------|
| <code>QL_UART_PORT_1</code>    | UART1           |
| <code>QL_UART_PORT_2</code>    | UART2           |
| <code>QL_UART_PORT_3</code>    | UART3           |
| <code>QL_USB_PORT_AT</code>    | USB AT port.    |
| <code>QL_USB_PORT_MODEM</code> | USB Modem port. |
| <code>QL_USB_PORT_NMEA</code>  | USB NMEA port.  |

### 2.3.1.2. `ql_uart_config_s`

The structure of serial port attribute configuration is defined as follows:

```
typedef struct
{
    ql_uart_baud_e baudrate;
    ql_uart_databit_e data_bit;
    ql_uart_stopbit_e stop_bit;
    ql_uart_paritybit_e parity_bit;
    ql_uart_flowctrl_e flow_ctrl;
}ql_uart_config_s
```

- Parameter

| Type                             | Parameter               | Description                                                                         |
|----------------------------------|-------------------------|-------------------------------------------------------------------------------------|
| <code>ql_uart_baud_e</code>      | <code>baudrate</code>   | Baud rate. The default value is 115200 bps. See <b>Chapter 2.3.1.4</b> for details. |
| <code>ql_uart_databit_e</code>   | <code>data_bit</code>   | Data bit. The default value is 8 bits. See <b>Chapter 2.3.1.5</b> for details.      |
| <code>ql_uart_stopbit_e</code>   | <code>stop_bit</code>   | Stop bit. The default value is 1 bit. See <b>Chapter 2.3.1.6</b> for details.       |
| <code>ql_uart_paritybit_e</code> | <code>parity_bit</code> | Parity bit. There is no parity by default. See <b>Chapter 2.3.1.7</b> for details.  |
| <code>ql_uart_flowctrl_e</code>  | <code>flow_ctrl</code>  | Flow control. It's disabled by default. See <b>Chapter 2.3.1.8</b> for details.     |

### 2.3.1.3. ql\_uart\_errcode\_e

The error codes of the serial port indicate whether the API execution is successful or not. If the function execution is failed, the reason for error is returned. The enumeration of error codes is defined as follows:

```
typedef enum
{
    QL_UART_SUCCESS                = 0,
    QL_UART_EXECUTE_ERR            = 1|QL_UART_ERRCODE_BASE,
    QL_UART_MEM_ADDR_NULL_ERR,
    QL_UART_INVALID_PARAM_ERR,
    QL_UART_OPEN_REPEAT_ERR,
    QL_UART_NOT_OPEN_ERR          = 5|QL_UART_ERRCODE_BASE,
} ql_uart_errcode_e
```

#### ● Member

| Member                    | Description                           |
|---------------------------|---------------------------------------|
| QL_UART_SUCCESS           | Successful execution.                 |
| QL_UART_EXECUTE_ERR       | Failed execution.                     |
| QL_UART_MEM_ADDR_NULL_ERR | The parameter address is NULL.        |
| QL_UART_INVALID_PARAM_ERR | Invalid parameter.                    |
| QL_UART_OPEN_REPEAT_ERR   | The serial port is opened repeatedly. |
| QL_UART_NOT_OPEN_ERR      | The serial port is not opened.        |

### 2.3.1.4. ql\_uart\_baud\_e

The enumeration of baud rate is defined as follows:

```
typedef enum
{
    QL_UART_BAUD_AUTO              = 0,
    QL_UART_BAUD_1200              = 1200,
    QL_UART_BAUD_2400              = 2400,
    QL_UART_BAUD_4800              = 4800,
    QL_UART_BAUD_9600              = 9600,
    QL_UART_BAUD_14400             = 14400,
    QL_UART_BAUD_19200             = 19200,
```

```

QL_UART_BAUD_28800    = 28800,
QL_UART_BAUD_33600    = 33600,
QL_UART_BAUD_38400    = 38400,
QL_UART_BAUD_57600    = 57600,
QL_UART_BAUD_115200   = 115200,
QL_UART_BAUD_230400   = 230400,
QL_UART_BAUD_460800   = 460800,
QL_UART_BAUD_921600   = 921600,
QL_UART_BAUD_1000000  = 1000000,
QL_UART_BAUD_1843200  = 1843200,
QL_UART_BAUD_2000000  = 2000000,
QL_UART_BAUD_2100000  = 2100000,
QL_UART_BAUD_3686400  = 3686400,
QL_UART_BAUD_4000000  = 4000000,
QL_UART_BAUD_4468750  = 4468750
}ql_uart_baud_e

```

#### ● Member

| Member                     | Description              |
|----------------------------|--------------------------|
| <i>QL_UART_BAUD_ATUO</i>   | Adaptive baud rate.      |
| <i>QL_UART_BAUD_1200</i>   | Baud rate is 1200 bps.   |
| <i>QL_UART_BAUD_2400</i>   | Baud rate is 2400 bps.   |
| <i>QL_UART_BAUD_4800</i>   | Baud rate is 4800 bps.   |
| <i>QL_UART_BAUD_9600</i>   | Baud rate is 9600 bps.   |
| <i>QL_UART_BAUD_14400</i>  | Baud rate is 14400 bps.  |
| <i>QL_UART_BAUD_19200</i>  | Baud rate is 19200 bps.  |
| <i>QL_UART_BAUD_28800</i>  | Baud rate is 28800 bps.  |
| <i>QL_UART_BAUD_33600</i>  | Baud rate is 33600 bps.  |
| <i>QL_UART_BAUD_38400</i>  | Baud rate is 38400 bps.  |
| <i>QL_UART_BAUD_57600</i>  | Baud rate is 57600 bps.  |
| <i>QL_UART_BAUD_115200</i> | Baud rate is 115200 bps. |
| <i>QL_UART_BAUD_187500</i> | Baud rate is 187500 bps. |
| <i>QL_UART_BAUD_230400</i> | Baud rate is 230400 bps. |

|                                   |                                                                  |
|-----------------------------------|------------------------------------------------------------------|
| <code>QL_UART_BAUD_460800</code>  | Baud rate is 460800 bps.                                         |
| <code>QL_UART_BAUD_921600</code>  | Baud rate is 921600 bps.                                         |
| <code>QL_UART_BAUD_1000000</code> | Baud rate is 1000000 bps.                                        |
| <code>QL_UART_BAUD_1843200</code> | Baud rate is 1843200 bps.                                        |
| <code>QL_UART_BAUD_2000000</code> | Baud rate is 2000000 bps. (Only supported by UART 2 and UART 3.) |
| <code>QL_UART_BAUD_2100000</code> | Baud rate is 2100000 bps.                                        |
| <code>QL_UART_BAUD_3686400</code> | Baud rate is 3686400 bps. (Only supported by UART 2 and UART 3.) |
| <code>QL_UART_BAUD_4000000</code> | Baud rate is 4000000 bps. (Only supported by UART 2 and UART 3.) |
| <code>QL_UART_BAUD_4468750</code> | Baud rate is 4468750 bps. (Only supported by UART 2 and UART 3.) |

#### 2.3.1.5. `ql_uart_databit_e`

The enumeration of data bit is defined as follows:

```
typedef enum
{
    QL_UART_DATABIT_7 = 7,
    QL_UART_DATABIT_8 = 8,
}ql_uart_databit_e
```

##### ● Member

| Member                         | Description                            |
|--------------------------------|----------------------------------------|
| <code>QL_UART_DATABIT_7</code> | 7 data bits (Not supported currently.) |
| <code>QL_UART_DATABIT_8</code> | 8 data bits.                           |

#### 2.3.1.6. `ql_uart_stopbit_e`

The enumeration of stop bit is defined as follows:

```
typedef enum
{
    QL_UART_STOP_1 = 1,
    QL_UART_STOP_2 = 2,
}ql_uart_stopbit_e
```

- Member

| Member                | Description  |
|-----------------------|--------------|
| <i>QL_UART_STOP_1</i> | 1 stop bit.  |
| <i>QL_UART_STOP_2</i> | 2 stop bits. |

### 2.3.1.7. ql\_uart\_parityit\_e

The enumeration of parity bit is defined as follows:

```
typedef enum
{
    QL_UART_PARITY_NONE,
    QL_UART_PARITY_ODD,
    QL_UART_PARITY_EVEN,
}ql_uart_paritybit_e
```

- Member

| Member                     | Description         |
|----------------------------|---------------------|
| <i>QL_UART_PARITY_NONE</i> | No parity bit.      |
| <i>QL_UART_PARITY_ODD</i>  | Enable odd parity.  |
| <i>QL_UART_PARITY_EVEN</i> | Enable even parity. |

### 2.3.1.8. ql\_uart\_flowctrl\_e

The enumeration of flow control is defined as follows:

```
typedef enum
{
    QL_FC_NONE = 0,
    QL_FC_HW,
}ql_uart_flowctrl_e
```

- Member

| Member                  | Description                    |
|-------------------------|--------------------------------|
| <code>QL_FC_NONE</code> | Disable hardware flow control. |
| <code>QL_FC_HW</code>   | Enable hardware flow control.  |

### 2.3.2. ql\_uart\_get\_dcbconfig

This function gets the attribute configuration of serial port.

- Prototype

```
ql_uart_errcode_e ql_uart_get_dcbconfig(ql_uart_port_number_e port, ql_uart_config_s *dcb)
```

- Parameter

*port*:

[In] Serial port number. See **Chapter 2.3.1.1** for details.

*dcb*:

[Out] Attribute configuration of serial port. See **Chapter 2.3.1.2** for details.

- Return Value

See **Chapter 2.3.1.3** for details.

### 2.3.3. ql\_uart\_open

This function opens the specified serial port.

- Prototype

```
ql_uart_errcode_e ql_uart_open(ql_uart_port_number_e port)
```

- Parameter

*port*:

[In] Serial port number. See **Chapter 2.3.1.1** for details.

- Return Value

See **Chapter 2.3.1.3** for details.

### 2.3.4. ql\_uart\_close

This function closes the specified serial port.

- **Prototype**

```
ql_uart_errcode_e ql_uart_close(ql_uart_port_number_e port)
```

- **Parameter**

*port:*

[In] Serial port number. See **Chapter 2.3.1.1** for details.

- **Return Value**

See **Chapter 2.3.1.3** for details.

### 2.3.5. ql\_uart\_write

This function writes data to the specified serial port.

- **Prototype**

```
int ql_uart_write(ql_uart_port_number_e port, unsigned char *data, unsigned int data_len)
```

- **Parameter**

*port:*

[In] Serial port number. See **Chapter 2.3.1.1** for details.

*data:*

[In] Data to be written.

*data\_len:*

[In] Length of the data to be written. Unit: byte.

- **Return Value**

Negative numbers      Failed execution. See **Chapter 2.3.1.3** for error codes.

Other values            Length of the actual written data.



### 2.3.6. ql\_uart\_read

This function reads the data from the specified serial port.

- **Prototype**

```
int ql_uart_read(ql_uart_port_number_e port, unsigned char *data, unsigned int data_len)
```

- **Parameter**

*port:*

[In] Serial port number. See **Chapter 2.3.1.1** for details.

*data:*

[Out] The data read.

*data\_len:*

[In] Length of the read data. Unit: byte.

- **Return Value**

Negative numbers      Failed execution. See error codes in **Chapter 2.3.1.3** for details.

Other values            Length of the actual read data.

### 2.3.7. ql\_uart\_register\_cb

This function registers the callback function for serial port events.

- **Prototype**

```
ql_uart_errcode_e ql_uart_register_cb(ql_uart_port_number_e port, ql_uart_callback uart_cb)
```

- **Parameter**

*port:*

[In] Serial port number. See **Chapter 2.3.1.1** for details.

*uart\_cb:*

[In] The callback function to be registered. If the function is set to NULL, the system is in polling pattern.  
See **Chapter 2.3.7.1** for details.

- **Return Value**

See **Chapter 2.3.1.3** for details.

### 2.3.7.1. ql\_uart\_callback

This function is the callback function of serial port events, used to notify the client to read the data.

- **Prototype**

```
typedef void (*ql_uart_callback)(uint32 ind_type, ql_uart_port_number_e port, uint32 size)
```

- **Parameter**

*ind\_type*:

[In] The types of serial port event. The events that can be handled on the App side include:

|                                             |                                                    |
|---------------------------------------------|----------------------------------------------------|
| <code>QUEC_UART_RX_RECV_DATA_IND</code>     | The event of receiving data.                       |
| <code>QUEC_UART_RX_OVERFLOW_IND</code>      | The event of buffer overflow after receiving data. |
| <code>QUEC_UART_TX_FIFO_COMPLETE_IND</code> | The event of finishing transmitting in FIFO mode.  |

*port*:

[In] The number of the serial port from which an event is sourced. See **Chapter 2.3.1.1** for details.

*size*:

[In] The size of data received by the serial port. Unit: byte.

- **Return Value**

None

# 3 Examples

This chapter introduces how to use the above API functions for serial port development and simple feature debugging on the App side. In the example, UART 2 is used for test by default.

## 3.1. Development Example

The example file *components\ql-application\peripheral\uart\_demo.c* of serial port API, is provided in the SDK codes of EC200U series and EG91xU family modules.

The file function is analyzed as follows:

- *ql\_uart\_app\_init()* //Creates a task. This function needs to be called before enabling the example.
- *ql\_uart\_demo\_thread()* //Executes the task. This function simply realizes the data configuring, opening, reading and writing of the serial port. The interrupt callback function reads the data by default.
- *ql\_uart\_notify\_cb()* //The callback function for serial port events, which simply realizes the data reading. The received data can be printed through coolwatcher.

If you need to enable this example, call *ql\_uart\_app\_init()* in *ql\_init\_demo\_thread()*, as shown in the following figure:

```

static void ql_init_demo_thread(void *param)
{
    QL_INIT_LOG("init demo thread enter, param 0x%x", param);

    ql_gpio_app_init();
    ql_gpioint_app_init();
    ql_ledcfg_app_init();

#ifdef QL_APP_FEATURE_AUDIO
    ql_audio_app_init();
#endif
#ifdef QL_APP_FEATURE_LCD
    ql_lcd_app_init();
#endif
    ql_nw_app_init();
    //ql_datacall_app_init();
    ql_osi_demo_init();

#ifdef QL_APP_FEATURE_FILE
    ql_fs_demo_init();
#endif

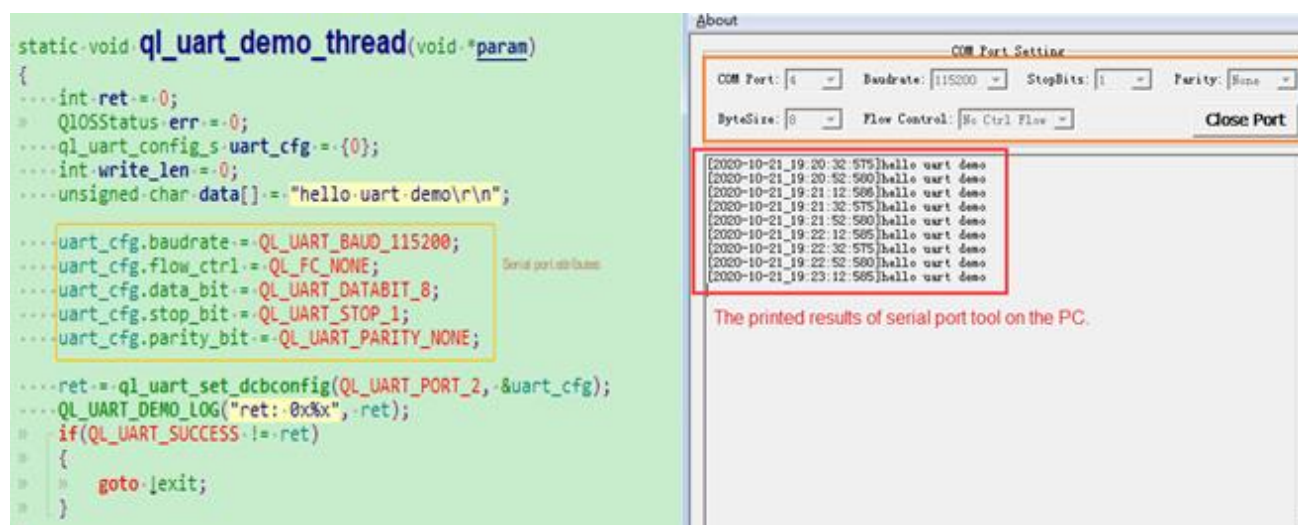
    //ql_dev_app_init();
    ql_adc_app_init();
    ql_uart_app_init();
#ifdef QL_APP_FEATURE_HTTP
    //ql_http_app_init();
#endif
}

```

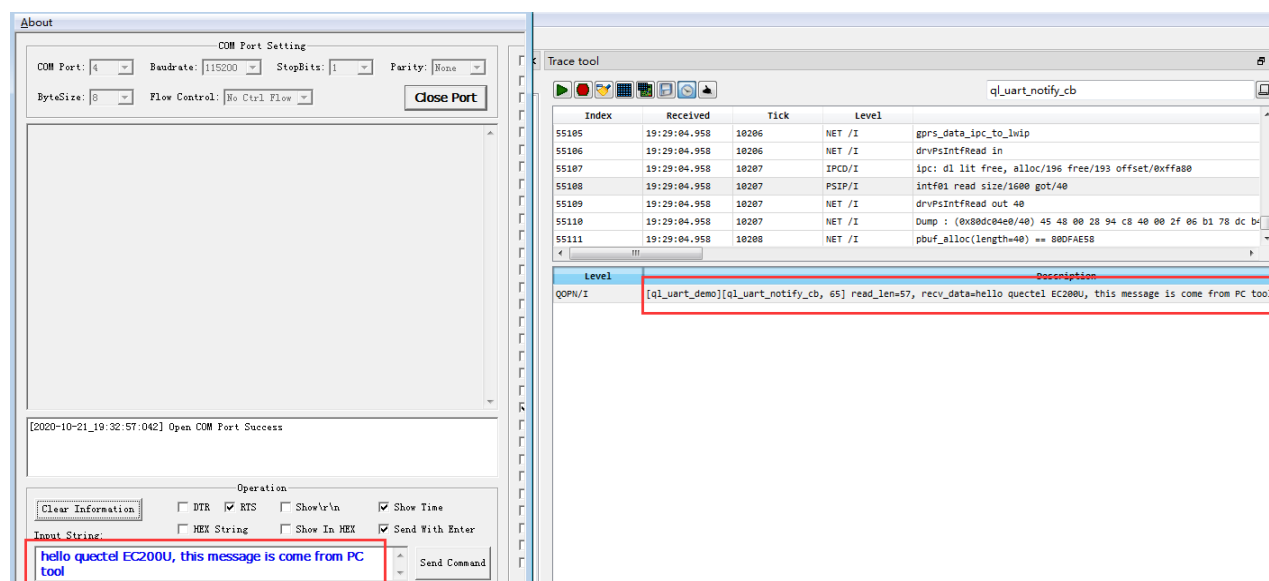
## 3.2. Feature Debugging

You can debug the serial port function of EC200U series and EG91xU family modules through LTE OPEN EVB.

Download the compiled firmware to the module and connect the USB port of the LTE OPEN EVB to the PC with a USB cable. The USB AP Log Port displays the system debugging information. See [document \[2\]](#) for log capture methods. Connect the cooltools tool, open the serial port tool (such as QCOM) on the PC and connect the serial port according to the serial port attributes set in the example. You can see that "hello uart demo" is received in the serial port tool of the PC as shown in the following figure:



Send the message "hello quectel EC200U series, this message is come from PC tool" through the serial port tool. The serial port data, received successfully by the module can be printed with cooltools, as shown in the following figure:



# 4 Appendix References

**Table 3: Related Documents**

| Document Name                                                    |
|------------------------------------------------------------------|
| [1] Quectel_EC200U&EG91xU_Series_QuecOpen(SDK)_Quick_Start_Guide |
| [2] Quectel_EC200U&EG91xU_Series_QuecOpen(SDK)_Log_Capture_Guide |

**Table 4: Terms and Abbreviations**

| Abbreviation | Description                                                            |
|--------------|------------------------------------------------------------------------|
| API          | Application Programming Interface                                      |
| App          | Application                                                            |
| EVB          | Evaluation Board                                                       |
| FIFO         | First In First Out                                                     |
| IoT          | Internet of Things                                                     |
| LTE          | Long-Term Evolution                                                    |
| NMEA         | NMEA (National Marine Electronics Association) 0183 Interface Standard |
| PC           | Personal Computer                                                      |
| RTOS         | Real-Time Operating System                                             |
| RX           | Receive                                                                |
| SDK          | Software Development Kit                                               |
| TX           | Transmit                                                               |
| DMA          | Direct Memory Access                                                   |